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Worksheet 1

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Aim:

Create a worksheet in Microsoft Excel containing student records. Demonstrate the use of the following functions: COUNT, COUNTA, COUNTBLANK, SUM, AVERAGE, IF, Nested IF, AND, OR, SUMIF, AVERAGEIF, and IFERROR.

Steps of Experiment:

1. Launch Microsoft Excel and open a blank workbook.
2. Create a table containing Roll Number, Student Name, Marks in Mathematics, Science, and English, Attendance Percentage, Sports Participation, and Remarks.
3. Enter details for at least ten students.
4. Calculate each student's total marks using the SUM function.
5. Find the average marks using the AVERAGE function.
6. Use the IF function to determine whether the student has passed or failed.
7. Apply Nested IF to assign grades according to the average marks.
8. Use the AND function to identify students meeting both attendance and academic requirements.
9. Apply the OR function to determine reward eligibility based on attendance or sports participation.
10. Count numerical values using COUNT.
11. Count filled cells using COUNTA.
12. Find empty cells with COUNTBLANK.
13. Calculate the total marks of students who passed using SUMIF.
14. Compute the average marks of passed students using AVERAGEIF.
15. Use IFERROR to display a custom message whenever an error occurs.
16. Verify all formulas and save the worksheet.

Experiment Solution

1.COUNT

Definition: Counts only the cells containing numeric values.
Example:
=COUNT(C2:C10)
Description: Returns the number of students whose marks have been entered.

```
="Total Students with Marks: " & COUNT(C2:C11)
```

Fig: Count Function

2.COUNTA

Definition: Counts all cells that are not empty.
Example: =COUNTA(B2:B11)
Explanation: Counts the total number of student names.

```
="Student Names Entered: " & COUNTA(B2:B11)
```

Fig:CountA Function

3.COUNTBLANK

Definition: The COUNTBLANK function counts the number of blank (empty) cells in a range. Example: =COUNTBLANK(L2:L11)

```
="Empty Remarks: " & COUNTBLANK(L2:L11)
```

Fig:CountBlank Function

4.SUM

Definition: The SUM function adds all the numeric values in a specified range.
Example:
=SUM(C6:E6)
Explanation: Calculates the Total Marks of a student.

```
="Total Marks of Passed Students: " & SUMIF(H2:H11,"Pass",F2:F11)
```

Fig:Sum Function

5.AVERAGE

Definition: The AVERAGE function calculates the arithmetic mean of a group of numbers.
Example:
=AVERAGE(C10:E10)
Explanation: Calculates the Average Marks of a student.

```
="Average Marks of Passed Students: " & ROUND(AVERAGEIF(H2:H11,"Pass",G2:G11),2)
```

Fig:Average Function

4.IF

Definition: The IF function checks a condition and returns one value if it is TRUE and another if it is FALSE.
Example:
=IF(G4>=40,"Pass","Fail")
Explanation: Displays Pass if the average is 40 or above; otherwise displays Fail.

```
="Safe Division Result: " & IFERROR(F2/J2,"Error")
```

Fig:IF Function

5.Nested IF

Definition: A Nested IF function uses multiple IF statements to test several conditions.
Example:
=IF(G3>=90,"A+",IF(G3>=75,"A",IF(G3>=60,"B",IF(G3>=40,"C","F"))))
Explanation: Assigns grades based on the student's average marks.

```
="Grade of " & B2 & ": " & IF(G2>=90,"A+",IF(G2>=75,"A",IF(G2>=60,"B",IF(G2>=40,"C","F"))))
```

Fig:Nested IF Function

6.AND

Definition: The AND function returns TRUE only if all specified conditions are true. Example:
=IF(AND(J5>=75,G5>=60),"Eligible","Not Eligible")
Explanation: Checks whether attendance is at least 75% and the average is at least 60.

```
="Eligibility of " & B2 & ": " & IF(AND(J2>=75,G2>=60),"Eligible","Not Eligible")
```

Fig:And Function

7.OR

Definition: The OR function returns TRUE if at least one of the specified conditions is true.
Example:
=IF(OR(J2>=90,K2="Yes"),"Reward","No Reward")
Explanation: Gives a reward if attendance is 90% or more or the student participated in sports.

```
="Reward Status of " & B2 & ": " & IF(OR(J2>=90,K2="Yes"),"Reward","No Reward")
```

Fig:OR Function

8.SUMIF

Definition: The SUMIF function adds values that satisfy a specified condition.
Example:
=SUMIF(H2:H11,"Pass",F2:F11)
Explanation: Calculates the total marks of all students who have Passed.

```
="Total Marks of Passed Students: " & SUMIF(H2:H11,"Pass",F2:F11)
```

Fig:Sumif Function

9.AVERAGEIF

Definition: The AVERAGEIF function calculates the average of values that satisfy a specified condition.

Example:

=AVERAGEIF(H2:H11,"Pass",G2:G11)

Explanation: Calculates the average marks of students who have Passed.

```
="Average Marks of Passed Students: " & ROUND(AVERAGEIF(H2:H11,"Pass",G2:G11),2)
```

Fig: Averageif Function

10.IFERROR

Definition: The IFERROR function returns a specified value if a formula results in an error.

Example:

=IFERROR(F2/J2,"Error")

Explanation: Displays "Error" instead of #DIV/0! if the attendance value is zero or blank.

```
="Division Result: " & IFERROR(F2/J2,"Error")
```

Fig:Iferror function

Output:

Roll No	Name	Math	Science	English	Total	Average	Result	Grade	Attendance	Sports	Remarks
101	Aarav Sharma	85	78	92	255	85	Pass	A	92	Yes	Eligible
102	Priya Verma	95	88	91	274	91.33	Pass	A+	96	Yes	Eligible
103	Rahul Singh	65	58	72	195	65	Pass	B	80	No	Eligible
104	Neha Gupta	55	60	57	172	57.33	Pass	C	70	No	Not Eligible
105	Mohit Kumar	75	82	79	236	78.67	Pass	A	90	Yes	Eligible
106	Anjali Patel	30	42	38	110	36.67	Fail	F	65	No	Not Eligible
107	Karan Mehta	88	90	84	262	87.33	Pass	A	94	Yes	Eligible
108	Sneha Joshi	68	70	74	212	70.67	Pass	B	88	No	Eligible
109	Riya Kapoor	99	95	97	291	97	Pass	A+	99	Yes	Eligible
110	Arjun Yadav	55	62	59	176	58.67	Pass	C	75	No	Not Eligible

Learning Outcome:

After completing this experiment, I was able to:

- Understand and apply basic and conditional Microsoft Excel functions such as SUM, AVERAGE, IF, AND, and OR for data analysis.
- Develop the ability to create and manage a student result worksheet using formulas and organized tabular data.
- Analyze and summarize data efficiently using functions like COUNT, COUNTA, COUNTBLANK, SUMIF, and AVERAGEIF.
- Implement logical decision-making in Excel by using Nested IF and IFERROR to automate grading and handle errors effectively.
- Enhance spreadsheet skills by performing calculations, validating results, and presenting data in a structured and professional format suitable for academic and business applications

Evaluation Grid :

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Worksheet Completion		08 marks
2.	Conduct		12 marks
3.	Viva Voice		10 marks
	Total		30 marks